

SAR Target Detection Based on SSD With Data Augmentation and Transfer Learning

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Abstract—In this letter, the single shot multibox detector (SSD), which is a real-time object detection method based on convolutional neural network, is applied to realize target detection for synthetic aperture radar (SAR) images. Since there are no sufficient labeled images for training in SAR target detection, we apply two strategies, data augmentation and transfer learning. For data augmentation, the first approaches to use some image processing methods, i.e., manual-extracting subimages, adding noise, filtering, and flipping, on the original training images to generate some new training images; the second approach is to employ the existing SAR target recognition data set, MSTAR data set, to assist in accomplishing the target detection task. For transfer learning, we first apply subaperture decomposition technique on original SAR images to acquire three-channel subaperture SAR images, and then transfer the three-channel VGGNet model pretrained on the ImageNet data set to the three-channel subaperture SAR images, in order to initialize corresponding parameters of the convolutional layers in the base network in our SSD. The feature extraction network, consisting of the base network and the auxiliary structure, is used to learn multiscale feature maps, and then convolutional predictors are used to acquire the final detection results. The experimental results on the miniSAR real image data set demonstrate that the proposed method can obtain better detection performance than other detection methods.

Index Terms—Convolutional neural networks (CNNs), data augmentation, single shot multibox detector (SSD), synthetic aperture radar (SAR), target detection, transfer learning.

I. INTRODUCTION

WITH the rapid development of the synthetic aperture radar (SAR) technology, the resolutions of SAR images are getting higher and higher, and more and more SAR images need to be processed. How to detect targets quickly and accurately from SAR images has become a research topic at present. The constant false alarm rate (CFAR) [1], which resorts to the pixel-level intensity differences between the targets and clutter, is commonly used for target detection in SAR images. A widely used CFAR is the two-parameter CFAR, which assumes the background clutter obeying the Gaussian distribution. The performance of the two-parameter CFAR is good in some simple scenes, but it may suffer from the performance degradation in some complex scenes.

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In recent years, convolutional neural networks (CNNs) have achieved great success in image classification, object detection, etc. At present, there are mainly two frameworks based on CNN for the object detection. The first is based on region proposals, such as R-CNN [2], [5], [6], [13] series. The second is based on the regression method, such as You Only Look Once (YOLO) [3] and single shot multibox detector (SSD) [4]. Due to the region proposal extraction step, R-CNN series cannot meet the real-time requirement. YOLO and SSD are real-time detection methods based on end-to-end deep neural networks. YOLO is faster than R-CNN series, but the detection accuracy is not so good, since only the features of the output layer are used. Compared with YOLO, SSD can further improve the detection accuracy via multiscale features.

CNN has achieved remarkable results in image classification and target detection, mainly owing to the massive labeled data sets in the field of natural images, such as the ImageNet [7] data set, which has roughly 1.2 million training images, and the PASCAL VOC2012 data set containing more than 20 000 training images. However, SAR images for training a good CNN model are insufficient. In the three-target MSTAR data set, only 698 training samples are commonly used for training. While in our original miniSAR data, only seven images are used for training and two images are used for testing. Obviously, the number of SAR images is significantly less than those in the ImageNet data set and PASCAL VOC2012 data set. Therefore, CNN cannot perform well with the insufficient training samples. To solve the problem of lacking training samples, references [8]–[10] adopted the data augmentation idea when applied CNN to realize SAR target recognition; Kang and He [11] adopted the idea of transfer learning when applied CNN to realize terrain classification for SAR images. For data augmentation, Ding *et al.* [8] used three methods, translation, adding noise, and pose synthesis; Chen *et al.* [9] applied the method of randomly sampling small patches; and Ødegaard *et al.* [10] just used simulated data as the new training samples. In the SAR target recognition task, training samples are usually small image chips cut from SAR images with large scenes, and each image chip usually has a single target and simple background. However, in the SAR target detection task, a training sample is usually an SAR image which may have multiple targets and complex background in a large scene. Therefore, some data augmentation methods in [8]–[10], which are designed for the target recognition task, are not suitable for the target detection task. For transfer learning, Kang and He [11] only mentioned that they adopted the method of direct parameters transfer, and did not specify how to transfer the CNN structure

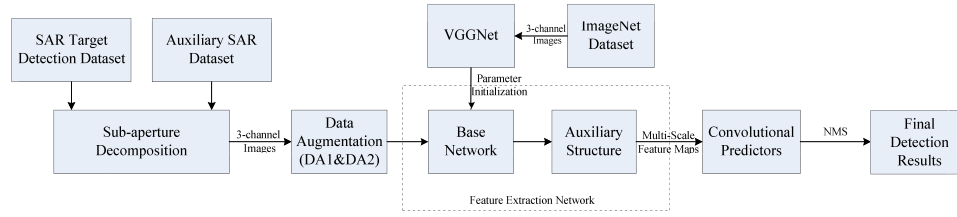


Fig. 1. Flowchart of the proposed SAR target detection method.

with the three channels for optical images to the one-channel SAR images.

As mentioned above, the existing data augmentation and transfer learning methods used in the SAR target recognition and terrain classification tasks are not suitable for the SAR target detection task. In this letter, we propose new data augmentation and transfer learning strategies for SAR target detection and apply the SSD detection network, which has good performance for optical images, to the SAR target detection task. For data augmentation, we apply two methods to generate new training samples. The first method is to operate on the original training images with some image processing methods; the second method is to pad MSTAR images with background pixels, where MSTAR images are commonly used for SAR target recognition. Here, we only choose some image processing-based data augmentation methods for SAR target detection problem by the detailed analysis of the existing methods. When using MSTAR data for simulation, we must first resize the MSTAR images and carefully control the ratio of simulated images from MSTAR data to original images from our detection data set. In order to transfer the well-learned network model on natural images to our SAR target detection application, since the colorful natural images have three channels, while the SAR images only have one channel, we should first transform the one-channel SAR images into three-channel images. To solve this problem, we utilize the subaperture decomposition technique [12] to obtain three-channel subaperture SAR images. Then, the three-channel VGGNet model pretrained on the ImageNet data set is transferred to the three-channel subaperture SAR images. Since the correlation properties among our three-channel SAR images are different from those among the three-channel colorful natural images, we just initialize the parameters of the convolutional layers in the base network of our SSD with the learned VGGNet model. The base network and the auxiliary structure are used to construct the feature extraction network for acquiring multiscale feature maps. For each feature map, a convolutional predictor is applied to get the detection result on each scale. The nonmaximum suppression (NMS) algorithm is later employed to remove redundant bounding boxes and get the final detection results. Our proposed method shows good detection performance based on the real SAR data.

II. PROPOSED TARGET DETECTION METHOD

The whole flowchart of the proposed detection method is given in Fig. 1. As shown in Fig. 1, the whole workflow of the proposed method is described as follows. First, we use the SAR target detection data set and the auxiliary SAR data set to generate the three-channel images via the subaperture

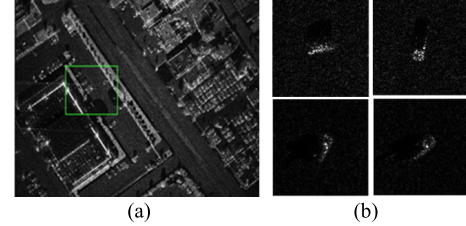


Fig. 2. Images from the miniSAR data set and the MSTAR data set. (a) MiniSAR image. (b) Some MSTAR images.

decomposition technology, which is a preprocessing step for transferring the three-channel VGGNet model to our SSD model. Second, we apply two data augmentation methods to the three-channel SAR images generated in the above to increase the number of training samples. Third, we initialize the parameters of the base network with the well-learned VGGNet model on the ImageNet data set, which realizes our transfer learning strategy. At the same time, the auxiliary structure is initialized randomly. Finally, the convolutional predictors and NMS algorithm are employed to obtain the final detection results.

A. Experimental Image Set Description

As shown in Fig. 1, we can see that the proposed detection method requires two different SAR data sets: SAR target detection data set and auxiliary SAR data set. In our experiments, we use the miniSAR data set as the target detection data set and the SAR target recognition data set, MSTAR data set, as the auxiliary data set to assist accomplishing the target detection task. The SAR images in the miniSAR data set contain many vehicle targets and complex background, including buildings, roads, grasslands, and trees. Their resolutions are $0.1 \text{ m} \times 0.1 \text{ m}$ and their sizes are 1638×2510 pixels. We choose nine SAR images that have vehicle targets from the miniSAR data set for our experiments and use seven images for training and the rest for testing. The MSTAR data set consists of multiple types of targets, such as BMP2, BTR70, and T72. The resolution for each MSTAR image is $0.3 \text{ m} \times 0.3 \text{ m}$, and the size is 128×128 pixels. Fig. 2(a) and (b) shows the SAR images from the miniSAR data set and MSTAR data set, respectively.

B. Subaperture Decomposing Technique

The colorful natural images have three channels, while the SAR images only have one channel. For optical images, to make full use of the data information, three-channel network structure is designed to describe the correlation among the three-channel data. Therefore, we also hope to generate multichannel data for the SAR images, which not only can make full use of the SAR data information but also can

Algorithm 1 Sub-Aperture Decomposition

Input: Original complex-valued SAR image $\mathbf{I}_{R \times A}$, where the vertical direction is the range dimension, and the horizontal direction is the azimuth dimension.

Output: 3 sub-aperture SAR images $\mathbf{I}_1, \mathbf{I}_2, \mathbf{I}_3$.

1. Implement 1-D Fourier transform along the azimuth direction of $\mathbf{I}_{R \times A}$ and obtain the azimuth spectrum $\mathbf{F}$.

2. Correct $\mathbf{F}$ to obtain the corrected azimuth spectrum $\mathbf{F}'$, by dividing the weighted function $\mathbf{W}$, which is estimated via the azimuth spectrum amplitude averaged along the range direction.

3. Divide $\mathbf{F}'$ into 3 non-overlapping sub-bands with equal length and obtain $\mathbf{F}'_1, \mathbf{F}'_2, \mathbf{F}'_3$,

4. Compute the weighted sub-bands $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3$, by multiplying $\mathbf{F}'_1, \mathbf{F}'_2, \mathbf{F}'_3$ with the weighted function $\mathbf{W}'$, where $\mathbf{W}'$ is obtained by downsampling $\mathbf{W}$ with 3 times.

5. Apply 1-D inverse Fourier transform to each sub-band along the azimuth direction, and obtain sub-aperture images $\mathbf{I}_1, \mathbf{I}_2, \mathbf{I}_3$.

realize the network transfer from the natural images to the SAR images. There are mainly two ways to convert the one-channel SAR image to three channels. One way is to duplicate the one-channel image into three channels, which is the same way as the Caffe [14] framework did. Here, the three-channel SAR images are the same, and they cannot provide more information for our SSD model. The other way is to use the subaperture decomposition technology [12] to obtain three subaperture images which correspond to the observations under different azimuth look angles. The targets usually have high scattering intensities in all the three subaperture images while the clutter does not have this characteristic. Thus, the three-channel subaperture SAR images may facilitate the SSD model to detect the targets from the clutter. Details of the subaperture decomposing technique [12] are listed in Algorithm 1.

C. Data Augmentation

In this letter, we use SSD based on CNN for SAR target detection, and training a good CNN model is important for SSD. Training a good CNN model needs a large number of training samples. However, in our SAR target detection task, our original training images are too insufficient to learn a good CNN model. In this letter, we propose two data augmentation methods suitable for SAR target detection, which are referred to as DA1 and DA2, respectively. Note that in the following, we only take one channel from the three-channel subaperture images as the example for the purpose of illustration.

1) *DA1*: We operate on the miniSAR subaperture images via four image processing approaches, i.e., manual-extracting subimages, adding noise, filtering, and flipping. We refer to this data augmentation method as DA1.

1) *Manual-Extracting Subimages*: As shown in Fig. 2(a), there exist vehicle targets and many kinds of clutter, including buildings and trees, in the SAR image. The vehicle targets are distributed relatively concentrated, and thus, we manually cut some training subimages

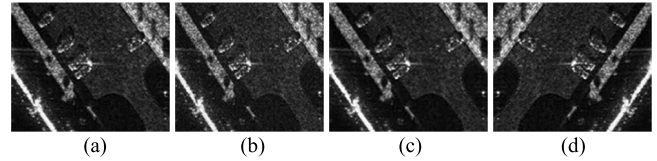


Fig. 3. Example of DA1. (a) Manual-extracted subimage. (b) Subimage added with the Gaussian noise. (c) Subimage filtered with the median filter. (d) Subimage flipped left to right.

containing the vehicle targets and typical clutter backgrounds from the original training images. Fig. 3(a) shows a subimage example cut from Fig. 2(a), which corresponds to the green rectangle region in Fig. 2(a). The size of the original miniSAR image is 1638×2510 , while the widths of extracted subimages are from 300 to 400, and the lengths are from 500 to 600.

- 2) *Adding Noise*: We add three kinds of noise, i.e., Gaussian noise, salt and pepper noise, and multiplicative noise, to the extracted subimages. Fig. 3(b) shows an example with adding the Gaussian noise.
- 3) *Filtering*: We employ two filtering methods, i.e., median filtering and Wiener filtering, to filter the extracted subimages. Fig. 3(c) shows an example with the median filtering.
- 4) *Flipping*: We flip the extracted subimages up to down and left to right. Fig. 3(d) shows an example, which is the left to right flipping result from Fig. 3(a).

In addition to the above four methods, randomly sampling small patches, translation, and pose synthesis are commonly used for data augmentation in SAR recognition application. As shown in Fig. 2(b), each image sample for SAR recognition usually only contains a single target and simple background. For randomly sampling small patches, the background should be uniform and the target should be at the center of the image. However, in the SAR image for target detection, as shown in Fig. 2(a), the targets are distributed randomly and the background is very complex. Therefore, we manually extract the subimages containing the targets and backgrounds, instead of randomly sampling small patches. The translation is an operation based on the centroid of a single target. As shown in Fig. 2(a), since there are usually multiple targets in the SAR images, the translation is not suitable for our target detection task. The pose synthesis requires some images of a single target from different poses and the corresponding pose information, which usually cannot be acquired for target detection task.

2) *DA2*: Since both of the image and target sizes are different for MSTAR images and miniSAR subimages, when using MSTAR images for data augmentation, we must adjust their image and target sizes similar to those of miniSAR subimages. Therefore, we first pad the MSTAR images with some background pixels, and then resize them. In addition, note that the MSTAR data set is just to assist accomplishing the target detection task on the miniSAR data set, and we should carefully control the ratio of simulated images from MSTAR data set to original images from miniSAR data set. Generally, the ratio should be about 2:1.

D. Feature Extraction Network

The feature extraction network consists of the base network and the auxiliary structure. Here, the base network [4] is modified based on the VGGNet. Here, we initialize the parameters of the convolutional layers in the base network of our SSD model with the well-learned VGGNet model on the ImageNet data set. The auxiliary structure is added to the end of the base network and it contains eight convolutional layers. In our SSD model for SAR target detection, the feature extraction network is employed to obtain multiscale feature maps from the bottom-level and top-level convolutional layers.

In this letter, we realize transfer learning via the following two steps: first, we use subaperture decomposing technique to acquire three-channel SAR images; second, we transfer the partial structure and initialize corresponding parameters from the VGGNet model for three-channel optical images to our SSD model for SAR target detection.

E. Convolutional Predictor

For each feature map which is obtained by the above feature extraction network, a convolutional predictor [4] is applied to get the detection result on each scale. Then, the NMS algorithm is employed to remove redundant bounding boxes among different scales to obtain the final detection results.

III. EXPERIMENTAL RESULTS

In this section, we validate the effectiveness of the proposed target detection method using the miniSAR data set, where vehicles are the targets of interest. The experiments are implemented with the Caffe [14] deep learning framework, using a personal computer with Intel Xeon E5-2630 v4 CPU of 2.2 GHz and NVIDIA GTX 1080Ti GPU on Ubuntu 16.04 Linux system.

A. Evaluation Criteria

We quantitatively evaluate different detection methods via precision, recall, and F1-score. They are defined as

$$\text{precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{recall} = \frac{TP}{NP} \quad (2)$$

$$F1 - \text{score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (3)$$

where TP is the number of really detected vehicle targets, FP is the number of false alarms, and NP is the number of total vehicle targets. F1-score is the harmonic mean between precision and recall, which is our main reference index to evaluate the detection performance.

B. Comparison Among SSD Models

To validate the effectiveness of the data augmentation and transfer learning methods for the proposed SAR target detection method, we compare seven SSD models, including our method and six variant versions, using miniSAR data set. These models are tested on two miniSAR images shown in Fig. 4, where the ground truths of the vehicle targets are marked by the green rectangles. We refer to the two images as

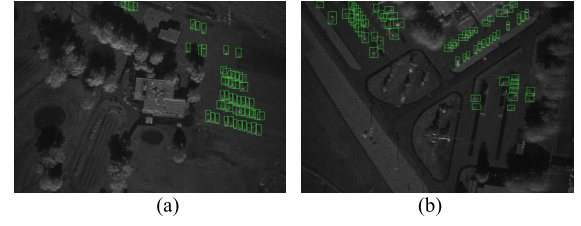


Fig. 4. Two testing images and their ground truths. (a) Image A. (b) Image B.

TABLE I
QUANTITATIVE COMPARISON AMONG DIFFERENT METHODS

Evaluation criteria Method	Precision (%)	Recall (%)	F1-score (%)
SSD	78.17 ± 2.00	90.52 ± 1.36	83.87 ± 1.05
SSD-DA1	83.86 ± 1.07	93.10 ± 0.61	88.24 ± 0.62
SSD-DA2	81.06 ± 0.97	92.93 ± 0.72	86.59 ± 0.55
SSD-DA1&DA2	84.72 ± 1.01	93.62 ± 0.47	88.95 ± 0.66
SSD-TL _{sub}	82.31 ± 0.72	94.66 ± 0.39	88.05 ± 0.48
SSD-DA1&2&TL _{dup}	85.43 ± 0.72	93.97 ± 0.61	89.49 ± 0.33
CFAR	63.69	86.21	73.26
Faster R-CNN	86.43 ± 2.25	90.69 ± 1.12	88.49 ± 0.88
The proposed method	87.01 ± 1.03	93.45 ± 0.47	90.11 ± 0.57

Image A and Image B, respectively. In test stage, we employ our trained SSD models to detect the vehicle targets via sliding window with overlapping in each test image. To eliminate the redundant bounding boxes introduced by overlapping between sliding windows, we employ NMS again to get the final detection results. Since there are some parameters need to be initialized randomly in the SSD-based deep neural network models, we implement each experiment five times to make their results more reliable.

Table I shows the quantitative comparison results among seven different SSD models. In Table I, SSD denotes the SSD without data augmentation and with random initialization; SSD-DA1 denotes the SSD with DA1 and random initialization; SSD-DA2 denotes the SSD with DA2 and random initialization; SSD-DA1&2 denotes the SSD with the proposed two data augmentation methods and random initialization; SSD-TL_{sub} denotes the SSD without data augmentation and with the proposed transfer learning method; and SSD-DA1&2&TL_{dup} denotes the SSD with the proposed two data augmentation methods and the transfer learning method which generates the three-channel images by duplicating the one-channel image into three channels.

As shown in Table I, SSD-DA1, SSD-DA2, and SSD-TL_{sub} perform better than SSD, which demonstrates that data augmentation or transfer learning is helpful for SSD to get better detection performance; SSD-DA1&2 presents better results than using SSD-DA1 and SSD-DA2 separately, which demonstrates that the two data augmentation methods are complementary to each other; and our proposed method, which considers both data augmentation and transfer learning, is the best. Moreover, as shown in Table I, the proposed method performs better than SSD-DA1&2&TL_{dup}, which demonstrates the effectiveness of the way of transforming the one-channel SAR images to the three-channel images by the subaperture decomposition technology.

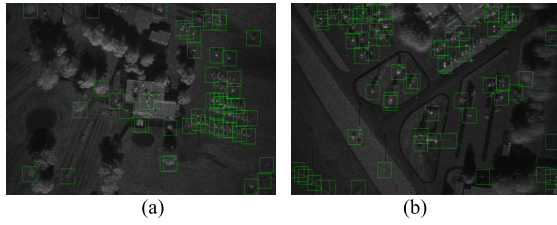


Fig. 5. Detection results of two-parameter CFAR. (a) Image A. (b) Image B.

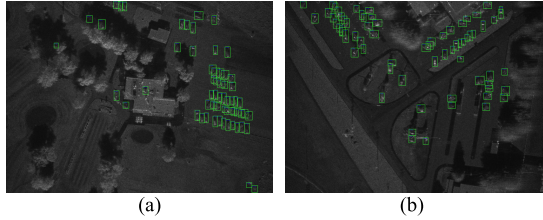


Fig. 6. Detection results of faster R-CNN. (a) Image A. (b) Image B.

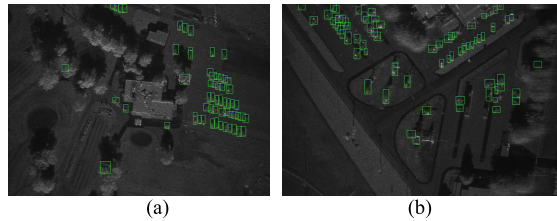


Fig. 7. Detection results of the proposed method. (a) Image A. (b) Image B.

C. Comparison With Other Detection Methods

To demonstrate the superior performance of the proposed detection method, we compare it with two-parameter CFAR and faster R-CNN. Since there are some parameters need to be initialized randomly in the faster R-CNN, we also implement the experiment five times. Nevertheless, the two-parameter CFAR does not need to be initialized randomly; thus, we only implement its experiment once. Table I gives the quantitative comparison among the three detection methods. Figs. 5–7 show the detailed detection results of three detection methods in one experiment.

As shown in Table I, the two-parameter CFAR method performs poorly. From Fig. 5, we can see that the two-parameter CFAR method has many false alarms and cannot locate the adjacent targets accurately. The main reasons are as follows. First, since CFAR only considers the intensity information, the false alarms are easy to be generated. Second, the clustering algorithm used in CFAR may merge two or more different targets into a target region. The CNN-based detection methods, including faster R-CNN and SSD, use not only the intensity information but also the structure information extracted by convolutional operators, which can reduce many false alarms. Moreover, the bounding box regressors are employed in the CNN-based detection methods to adjust the predicted bounding boxes, which can locate targets more accurately. It can also be seen from Table I that the proposed method based on SSD performs better than faster R-CNN. Faster R-CNN only uses single-scale feature maps from the output layer,

but SSD utilizes multiscale feature maps from the bottom and top convolutional layers. Bottom-level features reflect the information of contour edges in the targets, while top-level features reflect the semantic information of the targets, both of which are beneficial for the target detection task. We also compare the time costs between faster R-CNN and SSD on Image A and Image B: 35.94s for faster R-CNN and 20.32s for the proposed method. Since SSD uses the end-to-end deep neural network without the time-consuming region proposal extraction step, its speed is greatly improved.

IV. CONCLUSION

To deal with the problem of lacking training samples in SAR target detection, we utilize data augmentation and transfer learning strategies, and then apply the SSD method to accomplish the SAR target detection task. The experimental results based on the real SAR data set show that the proposed target detection method has much fewer false alarms and higher target positioning accuracy than the traditional CFAR detection method, which demonstrates the effectiveness of the proposed method. In the future work, we will consider replacing the VGGNet with the ResNet to see whether the detection results can be further improved.

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